FLUTIMATE 50

Fluticasone Pressurised Metered-Dose Inhaler 50mcg/dose (CFC-Free)

NAME AND STRUCTURE
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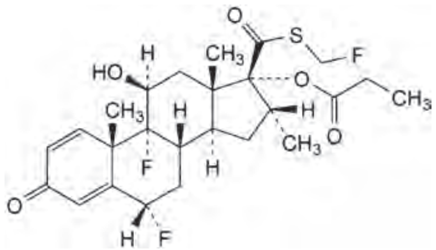
CHEMICAL STRUCTURE



MOLECULAR FORMULA $C_{25}H_{31}F_3O_5S$
MOLECULAR WEIGHT 500.6

PRODUCT DESCRIPTION

Fluticasone Propionate is a white or almost white suspension mixed with just right quantity of propellant and kept in a pressurised canister with a metered valve system.

PHARMACODYNAMIC

Fluticasone propionate given by inhalation at recommended doses has a potent glucocorticoid anti-inflammatory action within the lungs, resulting in reduced symptoms and exacerbations of asthma. There is a significant reduction of symptoms of chronic obstructive pulmonary disease (COPD) and an improvement in lung function regardless of patient age, gender, lung baseline function, smoking status or atopy status. This can result in a significant improvement in the quality of life.

PHARMACOKINETICS

The absolute bioavailability of inhaled fluticasone propionate varies between approximately 10-30% of the nominal dose depending on the inhalation device used. Systemic absorption occurs mainly through the lungs and is initially rapid then prolonged. The remainder of the inhaled dose may be swallowed but contributes minimally to systemic exposure due to the low aqueous solubility and presystemic metabolism, resulting in oral availability of <1%. There is a linear increase in systemic exposure with increasing inhaled dose. The disposition of fluticasone propionate is characterised by high plasma clearance (1150 mL/min), a large volume of distribution at steady state (approximately 300 L) and a terminal half-life of approximately 8 hrs. Plasma protein-binding is moderately high (91%). Fluticasone propionate is cleared very rapidly from the systemic circulation, principally by metabolism to an inactive carboxylic acid metabolite, by the cytochrome P-450 enzyme CYP3A4. The renal clearance of fluticasone propionate is negligible (<0.2%) and <5% as the metabolite. Care should be taken when co-administering known CYP3A4 inhibitors, as there is potential for increased systemic exposure to Fluticasone Pressurised Metered-Dose Inhaler 50mcg/dose (CFC-Free).

INDICATIONS

Adults: Prophylactic management in:

- Mild asthma (PEF values greater than 80% predicted at baseline with less than 20% variability): Patients requiring intermittent symptomatic bronchodilator asthma medication on more than an occasional basis.
- Moderate asthma (PEF values 60-80% predicted at baseline with 20-30% variability): Patients requiring regular asthma medication and patients with unstable or worsening asthma on currently available prophylactic therapy or bronchodilator alone.
- Severe asthma (PEF values less than 60% predicted at baseline with greater than 30% variability): Patients with severe chronic asthma. On introduction of inhaled fluticasone propionate many patients who are dependent on systemic corticosteroids for adequate control of symptoms may be able to reduce significantly or to eliminate their requirement for oral corticosteroids.

Children: Any child who requires preventative asthma medication, including patients not controlled on currently available prophylactic medication.

RECOMMENDED DOSE

Fluticasone Pressurised Metered-Dose Inhaler 50mcg/dose (CFC-Free) is administered by oral inhalation only.
Patients should be made aware of the prophylactic nature of therapy with inhaled Fluticasone Pressurised Metered-Dose Inhaler 50mcg/dose (CFC-Free) and that it should be taken regularly even when asymptomatic. The onset of therapeutic effect is 4-7 days, although some benefit may be apparent as soon as 24 hrs for patients who have not previously received inhaled steroids.
The dosage of fluticasone propionate should be adjusted according to the individual response.
If patients find that relief with short-acting bronchodilator treatment becomes less effective or they need more inhalations than usual, medical attention must be sought. It is intended that each prescribed dose is given by a minimum of 2 inhalations.
Asthma: -Adults and children over 16 years of age: 100 to 1000 micrograms twice daily. Patients should be given a starting dose of inhaled Fluticasone propionate which is appropriate for the severity of their disease: ***Mild asthma***: 100 to 250 micrograms twice daily. ***Moderate asthma***: 250 to 500 micrograms twice daily. ***Severe asthma***: 500 to 1000 micrograms twice daily.
The dose may then be adjusted until control is achieved or reduced to the minimum effective dose, according to the individual response.
Alternatively, the starting dose of fluticasone propionate may be gauged at half the total daily dose of beclomethasone dipropionate or equivalent as administered by metered-dose inhaler.
Children 4 years of age and over: 50 to 200 micrograms twice daily.
Many children's asthma will be well controlled using the 50 to 100 micrograms twice daily dosing regimen. For those patients whose asthma is not sufficiently controlled, additional benefit may be obtained by increasing the dose up to 200 micrograms twice daily.
Children should be given a starting dose of inhaled Fluticasone propionate that is appropriate for the severity of the disease.
The dose may then be adjusted until control is achieved, or reduced to the minimum effective dose, according to individual response.
Any child who requires prophylactic medication, including patients not controlled on currently available prophylactic medication. Once asthma symptoms have been controlled, the dose of fluticasone propionate should be reduced to the lowest dose which maintains control of asthma symptoms. The diagnosis and treatment of asthma should be kept under regular review.
Special patient groups: There is no need to adjust the dose in elderly patients or in those with hepatic or renal impairment.
Method of administration - Oral inhalation

CONTRAINDICATIONS

It is contraindicated in patients allergic reaction to any ingredient in the preparation.

WARNING AND PRECAUTIONS

The management of asthma should follow a stepwise programme, and patient response should be monitored clinically and by lung function tests. Increasing use of short-acting inhaled β2-agonists to control symptoms indicates deterioration of asthma control. Under these conditions, the patient's therapy plan should be reassessed.
Sudden and progressive deterioration in asthma control is potentially life threatening and consideration should be given to increasing corticosteroid dosage. In patients considered at risk, daily peak flow monitoring may be instituted.
Fluticasone Pressurised Metered-Dose Inhaler 50mcg/dose (CFC-Free) is not for use in acute attacks but for routine long-term management. Patients will require a fast- and short-acting inhaled bronchodilator to relieve acute asthmatic symptoms.
Lack of response or severe exacerbations of asthma should be treated by increasing the dose of inhaled fluticasone propionate and, if necessary, by giving a systemic steroid and/or an antibiotic if there is an infection.
Patients' inhaler technique should be checked to make sure that inhaler actuation is synchronised with inspiration to ensure optimum delivery of fluticasone propionate to the lungs.
Systemic effects may occur with any inhaled corticosteroid, particularly at high doses prescribed for long periods; these effects are much less likely to occur than with oral corticosteroids (see overdosage). Possible systemic effects include Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation in children and adolescents, decrease in bone mineral density, cataract and glaucoma. It is important, therefore, that the dose of inhaled corticosteroid is titrated to the lowest dose at which effective control is maintained (see Adverse Reactions).
It is recommended that the height of children receiving prolonged treatment with inhaled corticosteroid is regularly monitored.
Certain individuals can show greater susceptibility to the effects of inhaled corticosteroid than do most patients.

Adrenal function and adrenal reserve usually remain within the normal range on recommended doses of fluticasone propionate therapy. The benefits of inhaled fluticasone propionate therapy should minimise the need for oral steroids. However, the possibility of adverse effects in patients, resulting from prior or intermittent administration of oral steroids, may persist for some time. The extent of the adrenal impairment may require specialist advice before elective procedures. The possibility of impaired adrenal response should always be borne in mind in emergency and elective situations likely to produce stress and appropriate corticosteroid treatment must be considered (see Overdosage).
Because of the possibility of impaired adrenal response, patients transferring from oral steroid to inhaled fluticasone propionate therapy should be treated with special care and adrenocortical function regularly monitored.
Following introduction of inhaled fluticasone propionate, withdrawal of systemic therapy should be gradual and patients encouraged to carry a steroid warning card indicating the possible need for additional therapy in times of stress.
Similarly, replacement of systemic steroid treatment with inhaled therapy may unmask allergies eg, allergic rhinitis or eczema previously controlled by systemic drug. These allergies should be symptomatically treated with antihistamine and/or topical preparations, including topical steroids.
Treatment with Fluticasone Pressurised Metered-Dose Inhaler should not be stopped abruptly. There have been very rare reports of increases in blood glucose levels (see Adverse Reactions) and this should be considered when prescribing to patients with a history of diabetes mellitus.
As with all inhaled corticosteroids, special care is necessary in patients with active or quiescent pulmonary tuberculosis.
Effects on the Ability to Drive or Operate Machinery: Fluticasone propionate is unlikely to produce an effect.
Pneumonia in patients with COPD
An increase in the incidence of pneumonia, including pneumonia requiring hospitalisation, has been observed in patients with COPD receiving inhaled corticosteroids. There is some evidence of an increased risk of pneumonia with increasing steroid dose but this has not been demonstrated conclusively across all studies.
There is no conclusive clinical evidence for intra-class differences in the magnitude of the pneumonia risk among inhaled corticosteroid products.
Physicians should remain vigilant for the possible development of pneumonia in patient with COPD as the clinical features of such infections overlap with the symptoms of COPD exacerbations.
Risk factors for pneumonia in patients with COPD include current smoking status, older age, low body mass index (BMI) and severe COPD.

INTERACTIONS WITH OTHER MEDICAMENTS

Under normal circumstances, low plasma concentrations of fluticasone propionate are achieved after inhaled dosing, due to extensive first-pass metabolism and high systemic clearance mediated by cytochrome P-450 3A4 in the gut and liver. Hence, clinically significant drug interactions mediated by fluticasone propionate are unlikely.
Ritonavir (a highly potent cytochrome P-450 3A4 inhibitor) can greatly increase fluticasone propionate plasma concentrations, resulting in markedly reduced serum concentrations.
Other inhibitors of cytochrome P-450 3A4 produce negligible (erythromycin) and minor (ketoconazole) increases in systemic exposure to fluticasone propionate without notable reductions in serum cortisol concentrations. Nevertheless, care is advised when co-administering potent cytochrome P-450 3A4 inhibitors (eg, ketoconazole) as there is potential for increased systemic exposure to fluticasone propionate.
Co-treatment with CYP3A inhibitors, including cobicistat-containing products, it expected to increase the risk of systemic side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid side-effects.

PREGNANCY AND LACTATION

Pregnancy

There is inadequate evidence of safety of fluticasone propionate in human pregnancy. Tests of genotoxicity have shown no mutagenic potential. However, as with other drugs, administration of fluticasone propionate during pregnancy should only be considered if the expected benefit to the mother is greater than any possible risk to the foetus.

Lactation

The excretion of fluticasone propionate into human breast milk has not been investigated. However, plasma levels in patients following inhaled application of fluticasone propionate at recommended doses are likely to be low.

SIDE EFFECTS

Adverse events are listed as follows by system organ class and frequency. Incidence is defined as: very common, common, uncommon, rare, very rare, including case reports. Very common, common, less common are usually obtained by the clinical trial data. Rare and very rare adverse events are usually spontaneous information.

Infections and Infestations:

Very common: candidiasis of the mouth and throat.
Some patients will have oral and throat candidiasis (hungry Aphtha). Rinse mouth with water after treatment may be helpful to the patient. For symptoms of candidiasis, topical antifungal treatment may be useful and can continue to use the product.
Common: Pneumonia (in COPD patients)

Immune System Disorders:

Hypersensitivity reactions with the following manifestations have been reported: Uncommon: Cutaneous hypersensitivity reactions.
Very Rare: Angioedema (mainly facial and oropharyngeal oedema), respiratory symptoms (dyspnoea and/or bronchospasm) and anaphylactic reactions.

Endocrine Disorders:

Possible systemic effects include (see Precautions): Very Rare: Cushing's syndrome, Cushingoid features, adrenal suppression, growth retardation, decreased bone mineral density, cataract, glaucoma.

Metabolism and Nutrition Disorders:

Very rare: Hyperglycaemia.

Psychiatric Disorders:

Very rare: anxiety, sleep disorders, behavioral changes, including hyperactivity, irritability (mainly in children).

Respiratory, Thoracic and Mediastinal Disorders:

Common: Hoarseness.
In some patients, inhaled fluticasone propionate may cause hoarseness. It may be helpful to rinse out the mouth with water immediately after inhalation. Very Rare: Paradoxical bronchospasm.
As with other inhalation therapy, paradoxical bronchospasm may occur with an immediate increase in wheezing after dosing. This should be treated immediately with a fast-acting inhaled bronchodilator. Fluticasone Pressurised Metered-Dose Inhaler should be discontinued immediately, the patient assessed, and if necessary, alternative therapy instituted.

OVERDOSAGE

Acute inhalation of fluticasone propionate doses in excess of those approved may lead to temporary suppression of the hypothalamic-pituitary-adrenal axis. This does not usually require emergency action, as normal adrenal function typically recovers within a few days.
If higher than approved doses are continued over prolonged periods, significant adrenocortical suppression is possible. There have been very rare reports of acute adrenal crisis occurring in children exposed to higher than approved doses (typically $\geq$1000 mcg daily) over prolonged periods (several months or years); observed features included hypoglycaemia and sequelae of decreased consciousness and/or convulsions. Situations which could potentially trigger acute adrenal crisis include exposure to trauma, surgery, infection or any rapid reduction in dosage.
Patients receiving higher than approved doses should be managed closely and the dose reduced gradually.

PRESENTATION

120 doses per canister, contains 50mcg Fluticasone Propionate BP per actuation.

STORAGE

Store below 30°C. Protect from frost and direct sunlight.

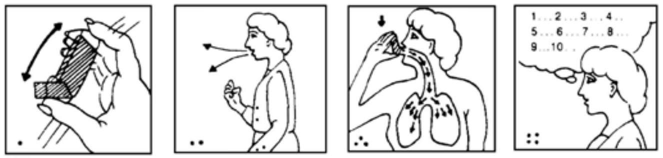
SHELF LIFE

24 months.

MANUFACTURER

Jewim Pharmaceutical (Shandong) Co., Ltd
Tai'an High-Tech Industrial Development Zone, Shandong Province, PRC.

HOW TO USE YOUR INHALER PROPERLY



1) Remove the cover from the mouthpiece and shake the inhaler vigorously.
2) Holding the inhaler as shown, breathe out gently (but not fully) and then immediately.
3) Place the mouthpiece in the mouth and close your lips around it. After starting to breathe in slowly and deeply through your mouth, press the inhaler firmly, as shown, to release it and continue to breathe in.
4) Hold your breath for 10 seconds, or as long as is comfortable, before breathing out slowly.
5) If you are to take a second inhalation you should wait at least one minute before repeating steps 2, 3 and 4.
6) After use, replace the cover on the mouthpiece.

Date of Revision: March 2020